

# ARDUINO - SMART OBSTACLE-AVOIDING ROVER

Explore Robotics, Ultrasonic Sensors & Autonomous Logic with Your Mini Robot!



## Required Materials for this Lesson:



1. Arduino Uno



2. HC-SR04 Sensor



3. L298N Module



4. 2× DC Motors



5. Robot Chassis



6. 9V Battery

## 1. How Does a Robot See & Decide?

Just like planetary rovers navigate alien terrain without human drivers, an autonomous obstacle-avoiding rover uses sensor feedback and software decision loops to detect hazards and dynamically adjust its path.

- **Arduino UNO (The Brain):** Microcontroller board that processes sensor feedback, calculates obstacle distances, and directs steering commands.
- **HC-SR04 Ultrasonic Sensor (The Eyes):** Emits high-frequency sound waves (40 kHz) and measures how long the echo takes to bounce back, working like echolocation in bats!
- **L298N Motor Driver (The Accelerator):** High-power dual H-bridge driver that amplifies low-power control signals from Arduino to drive dual DC motors.
- **2× DC Motors & Robot Chassis (The Muscle & Frame):** Converts electrical energy into dual-wheel steering and holds all hardware securely.

**⚠ ELECTRICAL SAFETY & MOTOR POWER WARNING!**  
**NEVER** power the DC motors directly from Arduino 5V or I/O pins. Always route current from the external 9V battery into the L298N power terminals (+12V and GND), and establish a shared GND connection between the Arduino and L298N module!

## 2. Complete Hardware Wiring Reference Table

Follow this master wiring chart to connect all components correctly across your chassis setup:

| From Component          | To Component              | Purpose / Description                           |
|-------------------------|---------------------------|-------------------------------------------------|
| HC-SR04 VCC             | Arduino 5V Pin            | 5V Power Supply for Ultrasonic Module           |
| HC-SR04 GND             | Arduino GND Pin           | Ground Return Path for Ultrasonic Sensor        |
| HC-SR04 Trig Pin        | Arduino Digital Pin 9     | Triggers Ultrasonic Sound Wave Pulses           |
| HC-SR04 Echo Pin        | Arduino Digital Pin 10    | Receives Ultrasonic Reflected Sound Echo        |
| L298N IN1 & IN2         | Arduino Pins 5 & 6        | Left Motor Direction Control (Forward/Reverse)  |
| L298N IN3 & IN4         | Arduino Pins 7 & 8        | Right Motor Direction Control (Forward/Reverse) |
| L298N OUT1 & OUT2       | Left DC Motor Terminals   | High Power Drives Left Wheel Motor              |
| L298N OUT3 & OUT4       | Right DC Motor Terminals  | High Power Drives Right Wheel Motor             |
| 9V Battery Positive (+) | L298N +12V Screw Terminal | Main Power Source for Both Motors               |
| 9V Battery Negative (-) | L298N GND Terminal        | Main Power Ground Connection                    |
| Arduino GND Pin         | L298N GND Terminal        | Common Ground Reference (Mandatory)             |

## 3. Detailed Step-by-Step Assembly & Wiring Steps

Follow these detailed steps to assemble your autonomous obstacle-avoiding rover:

### Step 1: Mechanical Chassis & DC Motor Mounting

Mount the two DC motors securely into the chassis cutouts using the provided brackets and screws. Attach the wheels onto the dual motor shafts. Securely bolt the Arduino Uno board and the L298N motor driver module onto the top platform of the chassis using standoffs or screws.

### Step 2: Connect DC Motors to L298N Terminals

Connect the lead wires from the **Left DC Motor** into screw terminals **OUT1** and **OUT2** on the L298N driver. Connect the lead wires from the **Right DC Motor** into screw terminals **OUT3** and **OUT4**. Tighten all screws firmly.

### Step 3: Wire L298N Control Inputs to Arduino

Using female-to-male jumper wires, connect **IN1** to Arduino Pin 5, **IN2** to Pin 6, **IN3** to Pin 7, and **IN4** to Pin 8. These pins dictate forward/reverse commands for both wheels independently.

### Step 4: Mount & Wire the HC-SR04 Ultrasonic Sensor

Attach the HC-SR04 sensor to the front bumper bracket of the chassis facing directly forward. Connect **VCC** to Arduino 5V, **GND** to Arduino GND, **Trig** to Digital Pin 9, and **Echo** to Digital Pin 10.

### Step 5: Connect 9V Battery & Common Ground

Connect the **Positive (+ / Red) wire** of the 9V battery clip to the **L298N +12V terminal**. Connect the **Negative (- / Black) wire** to the **L298N GND terminal**. Finally, run a wire from an **Arduino GND pin** into the same **L298N GND terminal** so both systems share a common ground reference.

## 4. Distance Calculation & Autonomous Logic

The ultrasonic sensor operates by transmitting a sound pulse and timing its reflection:

### Distance Calculation Formula:

$$\text{distance} = (\text{pulseDuration} \times \text{speedOfSound}) / 2$$

*Note: The total pulse duration is divided by 2 because the sound wave travels outward to the obstacle and bounces back to the sensor. Speed of sound in air  $\approx 0.0343$  cm/ $\mu$ s.*

### Rover Decision Code Example:

```
// Autonomous Rover Control Code
const int trigPin = 9;
const int echoPin = 10;

void setup() {
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  for (int i = 5; i <= 8; i++) { pinMode(i, OUTPUT); }
}

void loop() {
  digitalWrite(trigPin, LOW); delayMicroseconds(2);
  digitalWrite(trigPin, HIGH); delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  long duration = pulseIn(echoPin, HIGH);
  float distance = (duration * 0.0343) / 2;

  if (distance < 15) { // Obstacle detected within 15 cm threshold
    stopMoving();
    delay(300);
    turnRight();
    delay(500);
  } else {
    moveForward();
  }
}
```

### Threshold Distance Analysis:

- **Too Short (2 cm):** The rover moves too fast and collides before it has time to brake!
- **Just Right (15 cm):** Optimal safe stopping distance to evaluate surroundings and turn safely.
- **Too Long (50 cm):** The rover stops prematurely in open spaces long before reaching objects.

## 5. Testing & Troubleshooting Checklist

| Issue / Symptom                            | Troubleshooting Steps & Solution                                                                                                                                                                                                              |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Rover Moves Backward upon Sensing Obstacle | <ul style="list-style-type: none"><li>• Reverse motor wire connections at OUT1/OUT2 or OUT3/OUT4.</li><li>• Alternatively, flip HIGH/LOW pin logic in your code steering functions.</li></ul>                                                 |
| Distance Reads 0 cm or Unstable Values     | <ul style="list-style-type: none"><li>• Check Trig (Pin 9) and Echo (Pin 10) wiring.</li><li>• Verify 5V power and GND connections to the HC-SR04 sensor.</li><li>• Ensure 9V battery has sufficient voltage under load (&gt;7.5V).</li></ul> |
| One Wheel Fails to Spin                    | <ul style="list-style-type: none"><li>• Inspect L298N terminal screws for loose motor wire contact.</li><li>• Check Arduino digital output signals to IN1-IN4.</li></ul>                                                                      |

## Student Practice Sheet — Test Your Knowledge

Fill in your answers below or in your robotics lab notebook.

Name: \_\_\_\_\_ Date: \_\_\_\_\_

**Q1. Microcontroller Role**

What component serves as the "brain" of the obstacle-avoiding rover, and what is its primary role?

**Q2. Echolocation Principle**

Describe how the HC-SR04 Ultrasonic Sensor measures distance to an object in front of the rover.

**Q3. Mathematical Formula**

Why is the total pulse duration divided by 2 in the mathematical formula for calculating distance?

**Q4. Hardware Wiring Assignment**

Identify the default Arduino digital pins specified in the wiring guide for the Ultrasonic Trig pin and Echo pin.

**Q5. Threshold Distance Impact**

What will happen to the rover's behavior if the threshold distance in code is set to 2 cm instead of 15 cm?

**Q6. Steering Troubleshooting**

If your rover turns left instead of right when encountering an obstacle, what two steps can you take to fix it?

**Q7. Super Tinkerer Challenge**

Draw an obstacle course scenario and write the conditional decision steps (if/else) the rover should execute when approaching a wall.